

2-39 What is the range of R_{EQ} in Figure P2-39?

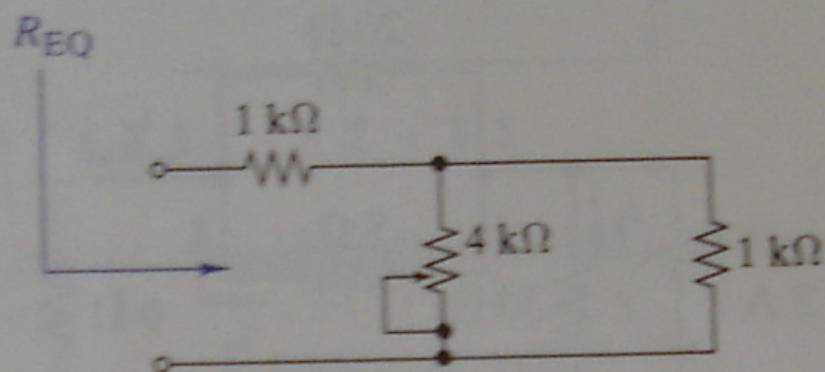


FIGURE P2-39

2-42 Use current division in Figure P2-42 to obtain an expression for v_L in terms of R , R_L , and i_S .

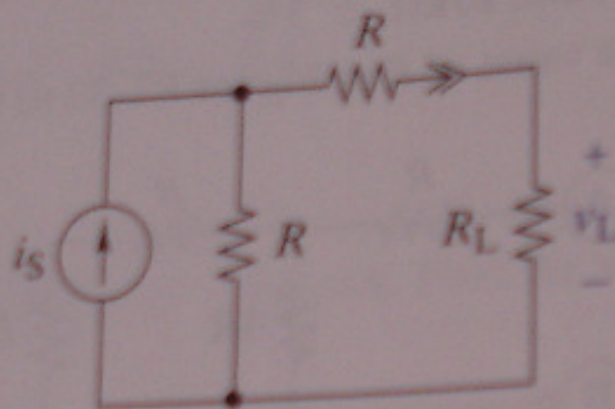


FIGURE P2-42

2-43 Find i_x in Figure P2-43.

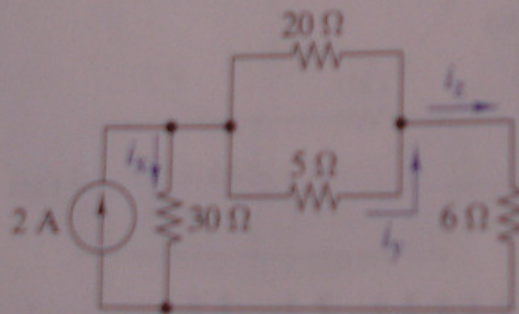


FIGURE P2-43

2-46 Figure P2-46 shows a resistance divider connected in a general circuit.

(a) What is the relationship between v_1 and v_2 when $i_1 = 0$?

(b) What is the relationship between v_1 and v_2 when $i_2 = 0$?

(c) What is the relationship between i_1 and i_2 when $v_1 = 0$?

(d) What is the relationship between i_1 and i_2 when $v_2 = 0$?

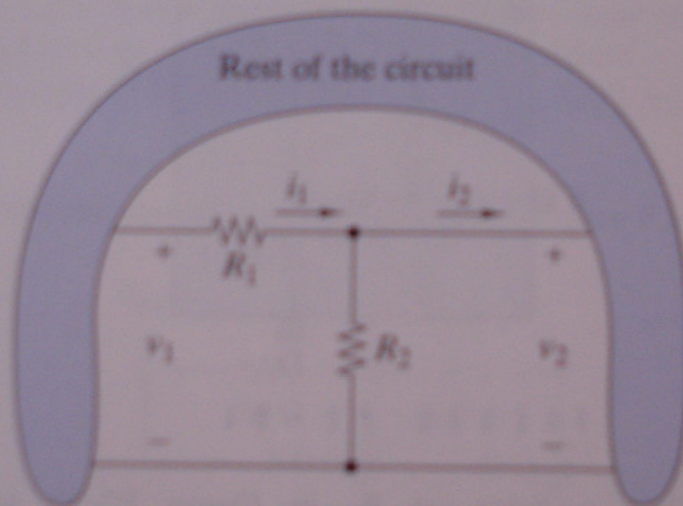


FIGURE P2-46

2-53 Use circuit reduction to find v_x in Figure P2-53.

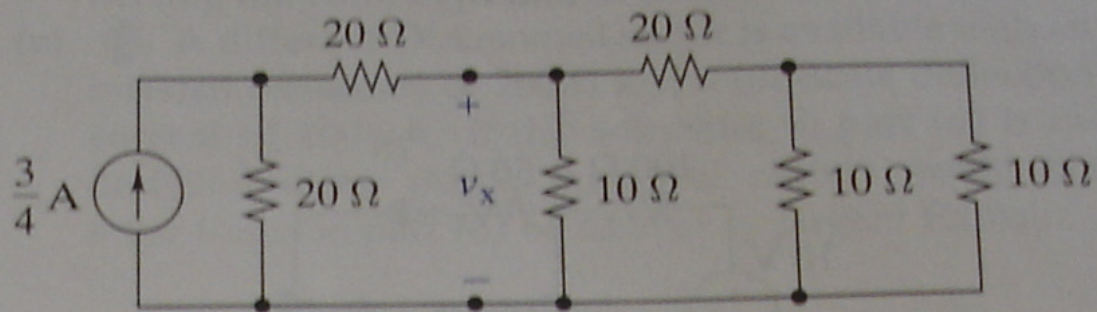


FIGURE P2-53

2-56 Use source transformation to find i_x in Figure P2-56.

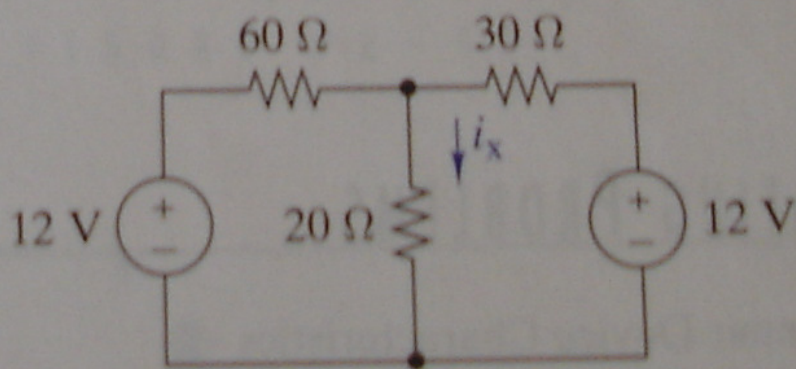


FIGURE P2-56

2-58 The current through R_L in Figure P2-58 is 40 mA. Use source transformations to find R_L .

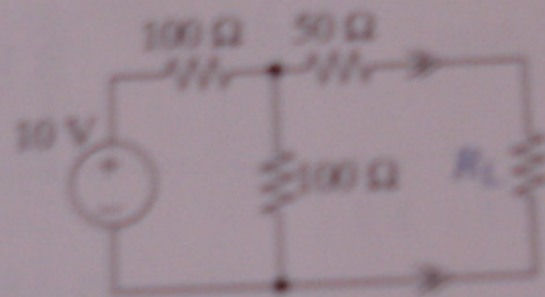


FIGURE P2-58

3-37 Find the Thévenin equivalent circuit seen by R_L in Figure P3-37. Find the voltage across the load when $R_L = 5\ \Omega$, $10\ \Omega$, and $50\ \Omega$.

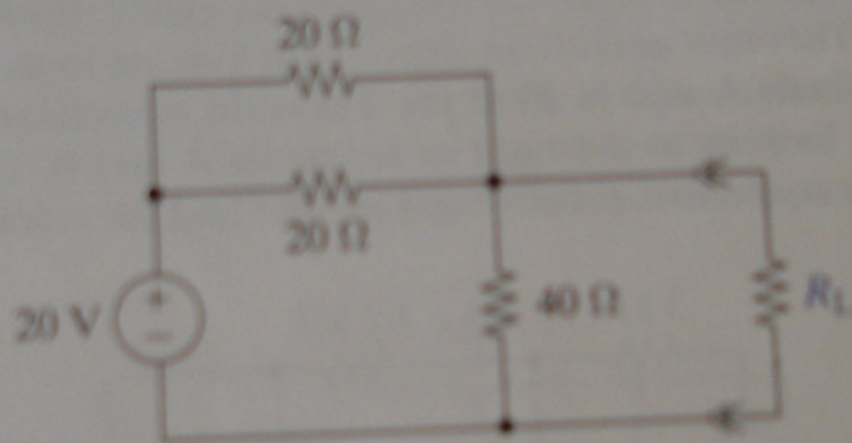


FIGURE P3-37

3-39 Find the Thévenin equivalent seen by R_L in Figure P3-39. Find the power delivered to the load when $R_L = 50\text{ k}\Omega$ and $200\text{ k}\Omega$.

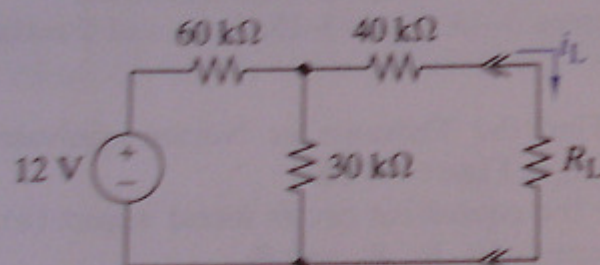


FIGURE P3-39

3-43 The i - v characteristic of the active circuit in P3-42 is $5v + 500i = 60$. Find the output voltage when a $500\text{-}\Omega$ resistive load is connected across the two accessible terminals.